

Centennial High School

Bitcoin: How the Industry Opened A Gateway to Criminal Activity

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Abstract:

The hypothesis of the impact that the Silk Road had on the fluctuations of Bitcoin prices was that throughout the Silk Road's active years; it directly led to Bitcoin's price rising 1,000% and after the shut down of the Silk Road; Bitcoin's price dropped by 200%. This hypothesis was theorized as the Silk Road played a pivotal role in Bitcoin's first 5 years as the Silk Road expanded Bitcoin's market and created opportunities for growth. The method of research chosen was the document analysis which allowed for comparisons and findings of direct correlations between the Silk Road and Bitcoin.

In the early 2010's the underground criminal organization, the Silk Road, organized its activities to be supported by the crypto currency Bitcoin. Bitcoin utilizes properties gained through the blockchain, including encrypted transactions, that allow it to evade the government's notice and protect its wallet holders (Dion-Schwarz 2019). As the Silk Road structured its market around Bitcoin; a new market and following was created for Bitcoin which allowed it to grow further and faster (Adler 2018). Bitcoin, with its newfound market began to attract government intervention and as its name began to be wide-spread its market grew yet again (Dion-Schwarz 2019). As Bitcoin and the Silk Road began to become deeper connected, fear grew among investors about whether the government would begin to take decisive action against the current (McMillan 2013). Although the Silk Road was shut down in 2013, its lasting impacts are viewed today as many crime syndicates and criminal organizations continue to abuse Bitcoin's anonymity in order to avoid government detection (Investopedia Team 2025). Bitcoin's current and past relations may bring uncertainty to investors and this essay aims to analyze the direct impact of the Silk Road on Bitcoin.

In the past decade, the financial field has transformed, with bitcoin leading the industry. This type of currency only exists online; it is not controlled by a central authority. Its value has rapidly increased, surpassing a \$2 trillion market cap in 2025. As a separate entity from government issued currency like the US dollar or euro, Bitcoin operates through a decentralized blockchain—a digital ledger responsible for recording transactions from computers—preventing control by a single entity (Dion-Schwarz 2019). Popularity in Bitcoin is gained by word of mouth; millions of dollars are invested daily in the hopes of making profit. Despite the rapid rise that Bitcoin is experiencing, the history remains clouded by controversy; illegal activity like

terrorist attacks and black markets happen largely due to the contributions of Bitcoin (Cryptocurrency Fraud Report 2023). Criminals have benefited from Bitcoin, using it as a way to take advantage of blockchains. For example, people have been able to conduct transactions without fear of government interference, regulation, control, tracking, and monitoring of financial activity (Adler 2018). Around the early 2010s, the expansion of Bitcoin became closely tied to the Silk Road through the creation of a new marketplace and surges in demand; allowing for Bitcoin to grow to new bounds (Harper 2020).

Bitcoin was created in 2009 by Satoshi Nakamoto, who was an anonymous user. Digital currency has brought attention to world wide investors, fluid and flowing in value unlike other currencies. Bitcoin runs on a blockchain, which uses ledgers in order to store data across different computers (Dion-Schwarz 2019). Bitcoin's concept, at the time, was unheard of as the currency was completely independent of the federal government and could not receive oversight. The rocky beginnings of Bitcoin are partially due to it being considered a hazard and a potential scam as investors would not understand the mechanics that play behind Bitcoin, causing it to take years of time for Bitcoin to be recognized as a form of safe investment that is unaffiliated with crime (Edwards 2024). Computers are used to mine Bitcoin pseudo-infinitely with the ceiling being at 25 million BTC (Investopedia 2025). Since Bitcoin's creation in 2009 it has continued to fluctuate in price greatly, sometimes with or without the crypto market following, creating uncertainty at times but still pulling through (Reed 2024).

The Silk Road originated as an unregulated and anonymous trading site that grew to infamy beginning in the 2010's. Founded by Ross William Ulbricht, it allowed buyers and sellers of illegal items to transact anonymously. The Silk Road allowed people to transact drugs, hacked passwords, illegal data, and other contraband; staying undetected as the website would only be

accessible through Tor, which functions as a digital black market which allows consumers to purchase illicit goods (Investopedia Team 2025). The FBI shut down Silk Road in 2013, and Ulbricht was sentenced to life in prison for his role (Investopedia Team 2025). Silk Road, Tor, and cryptocurrency were the ultimate privacy toolkits for illegal operations, providing a haven for drug traders. This marketplace sold a wide variety of goods and services that were non-existent in the surface level markets (Adler 2018). Emergence of the Silk Road emphasized the dangers of Bitcoin and solidified its standing in cyber-crime. As digital currency continues to pave its way into human history; it continues to raise demand for data privacy and concern over consumer safety, pushing the public into attempting to get restrictions placed on ways that sites like the Silk Road evade the federal government (Onion Sites 2025).

Bitcoin was primarily used in the Silk Road as a form of financial transaction that would allow the consumers and merchants to retain anonymity (Investopedia Team 2025). The Silk Road had sold countless services and goods that would be highly illegal in the retail above ground markets (Adler 2018). The selling of these goods would incur governmental investigation if even one transaction was compromised. As the Silk Road needed a method of transaction that would also completely isolate the Silk Road and keep the government in the dark; Bitcoin was chosen (Dion-Schwarz 2019). Bitcoin had exhibited all the traits that the Silk Road needed through the use of the blockchain. Bitcoin was implemented into the Silk Road and the Silk Road was coded to only receive and transmit transactions through the use of Bitcoin. The Silk Road would also continuously benefit from choosing Bitcoin as the price of Bitcoin rose which would account for more cash flow (Reed 2024). Bitcoin also drove foot-traffic to the Silk Road due to the Silk Road gaining infamy for being unregulated by the governmental authorities (due to Bitcoin's services). Bitcoin upheld the Silk Road's expectations as it continuously protected the

consumers and merchants from having their trades be compromised and getting sent to prison (Adler 2018).

Bitcoin had, in 2011, gained traction by the time the Silk Road was active. With Bitcoin's unique and distinct financial characteristics becoming more widely recognized, particularly its ability to operate beyond government regulation or oversight, the Silk Road saw an opportunity and made Bitcoin its primary method of financial exchange on the dark web, a hidden part of the internet used for anonymous transactions (Dion-Schwarz 2019). This allowed buyers and sellers to remain anonymous while keeping their transactions hidden from authorities, Bitcoin was able to avoid governmental financial regulation due to the blockchain which kept itself separate from government-imposed financial rules, where cryptocurrency exchanges, platforms that facilitate buying, selling, and trading digital currencies, take place (Dion-Schwarz 2019). Instead of personal identifiers, information that reveals the identity of a financial user, the blockchain relies on unique wallet addresses, alphanumeric codes representing digital wallets used for storing and sending cryptocurrency, which allow it to counter government tracing. While this didn't make the Silk Road completely immune to financial tracking, the ease of creating and discarding crypto wallets, digital accounts used for cryptocurrency storage and transactions, made tracking transactions significantly more difficult. This cemented the relationship between Bitcoin and the Silk Road, with Bitcoin serving as the backbone of the marketplace until authorities dismantled the Silk Road in late 2013 (McMillan 2013).

Bitcoin prices are constantly fluctuating according to the situations in the market and the real world. The most common factors that determine Bitcoin's prices are media influence, supply and demand, and government regulation. Unlike common company stocks where stock grants

partial ownership of a company, “buying bitcoin grants you ownership of however much cryptocurrency your money bought” (Bloomenthal 2024). Media influence plays a major role in shaping the price of Bitcoin as the media plays as a force of linkage between events surrounding Bitcoin and the investors. This factor both benefits and harms Bitcoin based on the information being relayed to the public. There have been many occasions where Bitcoin has gained large publicity, pushing more investors into investing into the crypto currency, or gained infamy, dragging away the current investors (Edwards 2024). An example of Bitcoin gaining rapid publicity is when the Silk Road was busted in 2013; this event transpired and led to the realization that without Bitcoin constantly keeping government officials at bay and hiding the transactions, it would have been impossible for the Silk Road to continue operating (McMillan 2013). Supply and demand plays the most crucial and constant role in Bitcoin’s price fluctuations. The supply and demand work similarly to as they do with company stocks where the lower the supply and higher the demand, the more driven the prices would be (Bloomenthal 2024). The other factors of Bitcoin always lead back to supply and demand as the more investors are lining up to buy Bitcoin, the more the demand goes up, leading to an inflated price. Supply and demand can work against Bitcoin when there is a surplus of supply but a lack of demand; Bitcoin suffered from this problem during its early months where the currency was not discovered by a wide audience, thus limiting the demand. As Bitcoin’s outreach was limited by its market, the currency took longer to grow. Governmental regulations play a detrimental role in shaping the future of Bitcoin as they drive prices drastically. Governmental regulations are often either restricting Bitcoin or giving it more freedom. The trend commonly indicates that as governmental restrictions increase, Bitcoin’s price drops. This happens as one of Bitcoin’s main alluring factors is its “freedom” and when that freedom is taken away, the market reacts.

Governmental regulations play into supply and demand as when Bitcoin is supported by a government, such as a government recognizing it as an official currency, the market for Bitcoin increases, leading to more demand for the currency and inflating the prices (Bloomenthal 2024).

The method of research chosen was the document analysis which allowed for comparisons and findings of direct correlations between the Silk Road and Bitcoin. The data collection method allowed for finding a direct correlation between Bitcoin and the Silk Road. The analysis of different sources would provide many accounts and also direct links between Bitcoin and the Silk Road. Through the analysis of charts that compare the increase of the price of Bitcoin on certain days compared to the events that positively reinforce the Silk Road, a correlation can be drawn between the two (Agarwal 2025).

Articles from sources such as Investopedia show that Bitcoin has seen volatile movements and instability throughout the time period of 2009 to 2013, this instability is assumed to be a byproduct of the relationship between Bitcoin and the Silk Road. The Silk Road created uncertain demand for Bitcoin and the prices were at times greatly inflated or deflated (Edwards 2024).

The article “How Silk Road Shaped Bitcoin's Evolution and Impact” by Raul Agarwal analyzes the actual interactions between Bitcoin and the Silk Road. The article states that the relationship between Bitcoin and the Silk Road has caused a shift in industries across the world, as Bitcoin and the Silk Road were able to affect policy agenda and create new industries (Agarwal 2025). The article by Alex Hern “Bitcoin price plummets after Silk Road closure” brings to light the true impacts that the closure of the Silk Road had on Bitcoin. At the time the current high of Bitcoin was at \$266 but immediately after the arrest of Ross Ulbricht and the

seizure of the site; Bitcoin's price plummeted (Hern 2025). This shows that Bitcoin has relied on the Silk Road and the Silk Road's market has been incorporated into Bitcoin's market.

This Chart shows the most impactful days that resulted in the gaining or loss of bitcoin prices and the activity that the Silk Road had partaken in during that time.

Date	Bitcoin Price (USD)	Silk Road Activity
Jan 2011	~\$0.30	Silk Road launches Feb 2011 (initial usage negligible)
Jun 2011 (first peak)	~\$30	Silk Road begins; volume still tiny
End 2012	~\$12.6	Silk Road trading ~6–9.5k BTC/day (≈\$15M/yr) ~4.5% of bitcoin trades
Apr 2013	~\$266	Silk Road activity high (mid-bull run)
Nov 2013 (pre-seizure)	~\$1,100	Silk Road total sales ≈9.5M BTC (≈\$1.2B) ; ~20%

		of BTC “economic activity” at peak
Oct 2013 (closure)	~\$204	Silk Road site seized and closed by FBI (~26k BTC seized)

The direct impacts that the Silk Road had on Bitcoin is inferred to be 5-15% of all Bitcoin price movements in the period of 2011-2013. Although the Silk Road may have been a large contributing factor from below the market; Bitcoin above the market was rapidly rallying support and growing in size (Agarwal 2025). The criminal enterprise would only account for 15 percent of trades at most and 5 percent at least in the allotted time period. As Bitcoin had risen from 30 cents to over \$1,100, the percent that it gained in the time period is roughly 366,567%, if the Silk Road had a 9% impact in the overall; it would bring their direct percentage of influence to 32,991.03%. Immediately after the shutdown of the Silk Road, Bitcoin prices plummeted to roughly \$200, the result shows that the price dropped by over 81.45% and as the event took place immediately after Bitcoin’s closing; it can be assumed that they are directly and almost completely related (McMillan 2013).

Limitations of the research conducted include that there is no real and completely accurate method in order to judge the impact that the Silk Road has on Bitcoin. Bitcoin is a currency that can be affected by countless factors and these factors each have their own methods

of influencing Bitcoin prices, these influences may reinforce the impact the Silk Road had on Bitcoin or go against it, making its impact less profound (Cryptocurrency Fraud Report 2023). As these external factors are pseudo-infinite, there is no real way to gauge the impact, and only educated assumptions can be created.

If the research process was not interrupted I would be able to continue to isolate the Silk Road's impact on Bitcoin and attempt to get the closest possible solution. In order to only isolate the influence of the Silk Road, the other influences must be removed, and in order to remove those influences the relative impact that they had on Bitcoin needs to be estimated. After the estimations are done, it needs to be reflected on the overall impact and removed, after this process is repeated countless times, the Silk Road's impact will be found. This process has its cons as it may be inaccurate as the estimate for impacts of each influence may be distorted.

Bitcoin and the Silk Road have had ties since both of their creations, leading to a propelling of both forces. The Silk Road assisted and harmed Bitcoin through their interactions. The Silk Road may have assisted Bitcoin in finding its initial market but it also was a criminal market, leading to casual investors being wary of the currency. The Silk Road would have also led to Bitcoin being more regulated by the federal government as it would be associated with a large criminal enterprise (Dion-Schwarz 2019). Understanding Bitcoin's past relations is crucial for investors that look into Bitcoin. Bitcoin has shown that its past and current connections with the Silk Road could suggest a restriction on the blockchain as it openly promotes criminal activity. Bitcoin's price has been directly shown to be massively impacted by the incidents in the Dark web, suggesting unreliability. During the early 2010's, Bitcoin's growth is directly attributed to the Silk Road through the creation of a foundation, new market, and an increase in demand (Harper 2020).

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